REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

How Fast Practice Questions

Grade 9 Advanced | Module 2 Lesson 4 | How Fast

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: $v_{avg} = \Delta x/\Delta t$; average speed = total distance/total time; $\Delta x = v\Delta t$ for constant velocity.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 42–48.

P01. Average velocity is calculated using:

A. Displacement divided by time interval B. Distance multiplied by time
C. Speed divided by distance D. Final position alone

P02. Which quantity is always nonnegative?

A. One-dimensional velocity B. Speed
C. Displacement D. Position coordinate

P03. Instantaneous velocity is represented on a position–time graph by:

A. Total graph height divided by mass B. Area under the graph
C. Slope of the tangent at an instant D. The time-axis intercept alone

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P04. A round trip can have:

A. Negative total distance B. Nonzero displacement with identical endpoints
C. Zero travel time D. Zero average velocity but nonzero average speed

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P05. Uniform velocity means constant:

A. Speed and direction B. Speed only with any direction
C. Position D. Acceleration of any value

P06. An object moves from -6 m to 12 m in 8 s. Calculate its average velocity.

A. 2.25 m/s B. 4.5 m/s
C. 1.125 m/s D. -2.25 m/s

P07. A runner covers 12 m outward and 8 m back in 10 s. Calculate the average speed.

A. 4 m/s B. 2 m/s
C. 1 m/s D. -2 m/s

P08. An object moves at constant velocity +6 m/s for 6 s. Calculate its displacement.

A. 72 m B. 18 m
C. 36 m D. -36 m

P09. A cyclist travels 120 m at constant speed 10 m/s. Calculate the travel time.

A. 24 s B. 6 s
C. -12 s D. 12 s

P10. Convert a speed of 72 km/h to m/s.

A. 20 m/s B. 40 m/s
C. 10 m/s D. -20 m/s

P11. Distinguish average speed from average velocity.

P12. Define instantaneous velocity.

P13. When does average speed equal the magnitude of average velocity?

P14. What does a negative average velocity mean in one dimension?

P15. Why must total time include stops when calculating overall average speed?

P16. An object moves from -9 m to 18 m in 12 s. Calculate its average velocity.

P17. An object moves from -12 m to 24 m in 16 s. Calculate its average velocity.

P18. A runner covers 18 m outward and 12 m back in 15 s. Calculate the average speed.

P19. A runner covers 24 m outward and 16 m back in 20 s. Calculate the average speed.

P20. An object moves at constant velocity +9 m/s for 7 s. Calculate its displacement.

P21. An object moves at constant velocity +12 m/s for 8 s. Calculate its displacement.

P22. A cyclist travels 180 m at constant speed 15 m/s. Calculate the travel time.

P23. A cyclist travels 240 m at constant speed 20 m/s. Calculate the travel time.

P24. Convert a speed of 108 km/h to m/s.

P25. Convert a speed of 144 km/h to m/s.

P26. A runner travels 100 m east in 20 s and returns 100 m west in 30 s. Find overall average speed and average velocity.

P27. A student averages two speeds, 4 m/s and 8 m/s, for two equal-distance legs. Is 6 m/s the trip average? Explain.

P28. A position–time curve is tangent to a horizontal line at one instant. What is the instantaneous velocity then?

P29. Explain why a speedometer reading does not necessarily equal the average speed of a long trip.

P30. Design a simple method to measure a student's average walking speed along a straight corridor.